TECHNOLOGICAL CHINA / PPT 5

ELECTRIC VEHICLES, BATTERIES AND SMART FACTORIES

China and the new industrial revolution

Sergio Gevatschnaider



One transition, four transformations

Convergence matters more than each technology in isolation.

01

Electrification

Motors, power electronics and charging networks.

02

Chemistry

Cells, materials, costs, degradation and recycling.

03

Digitalisation

Software, sensors, data and AI in vehicles and plants.

04

Industry

Scale, suppliers, automation and learning.

Thesis: competitive advantage comes from the system, beyond any single product.

Three measures of the transformation

Different indicators with different denominators.

≈55%

Electric share of new car sales in China

2025 sales share · IEA

>80%

Battery cells produced in China

2025 global production share · IEA

354,000

Industrial robots installed in China

2025 annual installations · IFR

Distinguish sales share, cell production and annual robot installations.

SOURCE IEA

The economic question: where is value created?

From minerals to software, from products to data.

01

Materials

Refined minerals and chemistry.

→

02

Cell

Electrodes, safety and performance.

→

03

Vehicle

Design, electronics and production.

→

04

Service

Software, charging and after-sales.

Value capture depends on complementary capabilities as well as volume.

MARKET

01 · MARKET AND TRANSITION

From conventional motoring to mass electrification



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China’s market: scale changes product economics

IEA electric car sales include BEVs and PHEVs.

>13 M

Electric cars sold in China

2025 · IEA

≈55%

Share of new car sales

2025 · IEA

≈44 M

Electric cars on the road

End of 2025 · IEA

Scale economies, learning and domestic competition shape price and variety.

SOURCE IEA

BEVs and PHEVs: two different economic architectures

Avoid equating the two technologies across all indicators.

BEV · battery electric

- No combustion engine for propulsion.
- Charging infrastructure is crucial.
- Battery and software are major cost drivers.
- Range and chemistry shape performance.

PHEV · plug-in hybrid

- Combines fuel with external charging.
- Less range anxiety, heterogeneous usage.
- Two powertrains increase complexity.
- Real emissions depend on charging frequency.

A rigorous comparison defines use patterns, electricity mix and vehicle segment.

Four drivers of adoption in China

Supply, demand and institutions interact.

01

Prices

Competition, scale and cost improvements.

02

Variety

Urban models, sedans, SUVs and commercial vehicles.

03

Infrastructure

Charging, power grids and charging services.

04

Policy

Standards, incentives and local regulation.

Markets expand when total cost, convenience and supply align.

BYD: from battery maker to industrial platform

NEVs include BEVs and PHEVs, beyond pure battery electric vehicles.

4.6 M

New energy vehicles sold

BYD · 2025

RMB 804 bn

Corporate revenue

BYD · 2025

>1 M

Units sold outside China

BYD · 2025

Vertical integration improves coordination while concentrating capital risks.

SOURCE BYD

A price war does not establish efficiency

What to evaluate beyond market share.

Consumer gains

- Access to more affordable models.
- Rapid improvements in performance.
- Innovation pressure on manufacturers.
- Diffusion into new markets.

Adjustment costs

- Lower margins and idle capacity.
- Financial risks for suppliers.
- Pressure on quality and profitability.
- Trade tensions and subsidies.

Measure productivity, profitability and survival alongside sales.

BATTERIES

02 · BATTERIES AND MATERIALS

The cell at the heart of industrial competitiveness



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From electrodes to battery pack

A simplified cell-and-pack manufacturing process.

01

Electrodes

Mixing, coating and drying.

→

02

Cell assembly

Separator, stacking and electrolyte.

→

03

Formation

Initial charging, ageing and grading.

→

04

Pack integration

Cell-to-pack or modules, BMS and tests.

Production yield and traceability affect final cost.

LFP versus NMC: a multidimensional frontier

No chemistry is superior in every application.

LFP · lithium iron phosphate

- Lower material cost, without nickel or cobalt.
- Good thermal stability; favourable safety profile.
- Typically lower energy density than NMC.
- Attractive for volume markets and storage.

NMC · nickel manganese cobalt

- Higher energy density in many designs.
- Useful for long-range vehicles.
- Higher costs and mineral exposure.
- Needs thermal and degradation management.

Compare total cost, density, safety, lifetime, climate and recyclability.

The growing role of LFP batteries

Shares and prices refer to different populations.

>55%

LFP share of deployed EV batteries

Global · 2025 · IEA

>40%

Lower average LFP price than NMC

Packs per kWh · 2025 · IEA

>80%

Cells manufactured in China

Global production share · 2025 · IEA

Average pack prices include storage applications; this is not a controlled chemistry test.

SOURCE BAT

Pack anatomy: from electrode to vehicle

Components that determine real-world performance.

01

Cell

Chemistry, capacity, temperature and degradation.

02

BMS

SOC/SOH estimation and protection.

03

Thermal system

Cooling, heating and safety.

04

Structure

Cell-to-pack and cell-to-body integration.

System design matters alongside chemistry.

CATL: scale, innovation and reach

Company-reported figures from its annual report.

661 GWh

Lithium battery sales

2025 · CATL

772 GWh

Global production capacity

2025 · CATL

RMB 22.1 bn

Research and development spending

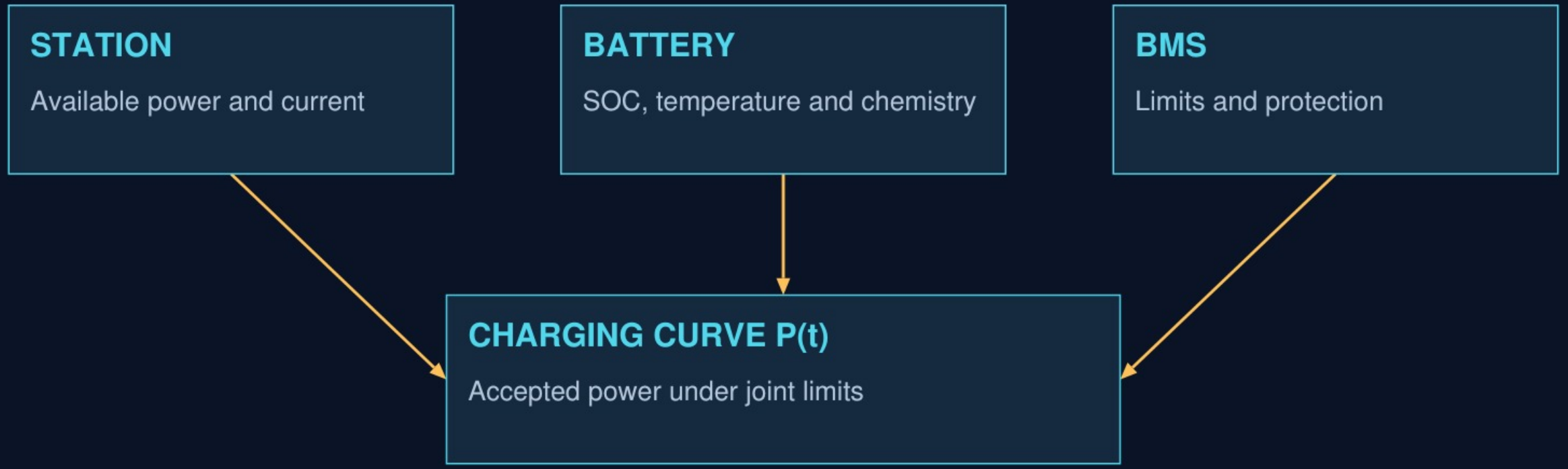
2025 · CATL

Comparing GWh sold with vehicles requires an explicit average pack size.

SOURCE CATL

Fast charging: what does it actually mean?

Physics and constraints of charging.



P(t) = V(t) × I(t) · Time ≈ required energy / average power

New technological pathways: progress and limits

Assess industrial maturity before extrapolating.

01

Sodium-ion

An alternative for selected uses and costs.

02

Solid-state

Potential gains, manufacturing challenges.

03

LMFP

Trade-offs between cost and density.

04

Recycling

Material recovery and traceability.

Pilot trials ≠ industrial scale ≠ sustained competitiveness.

MANUFACTURING

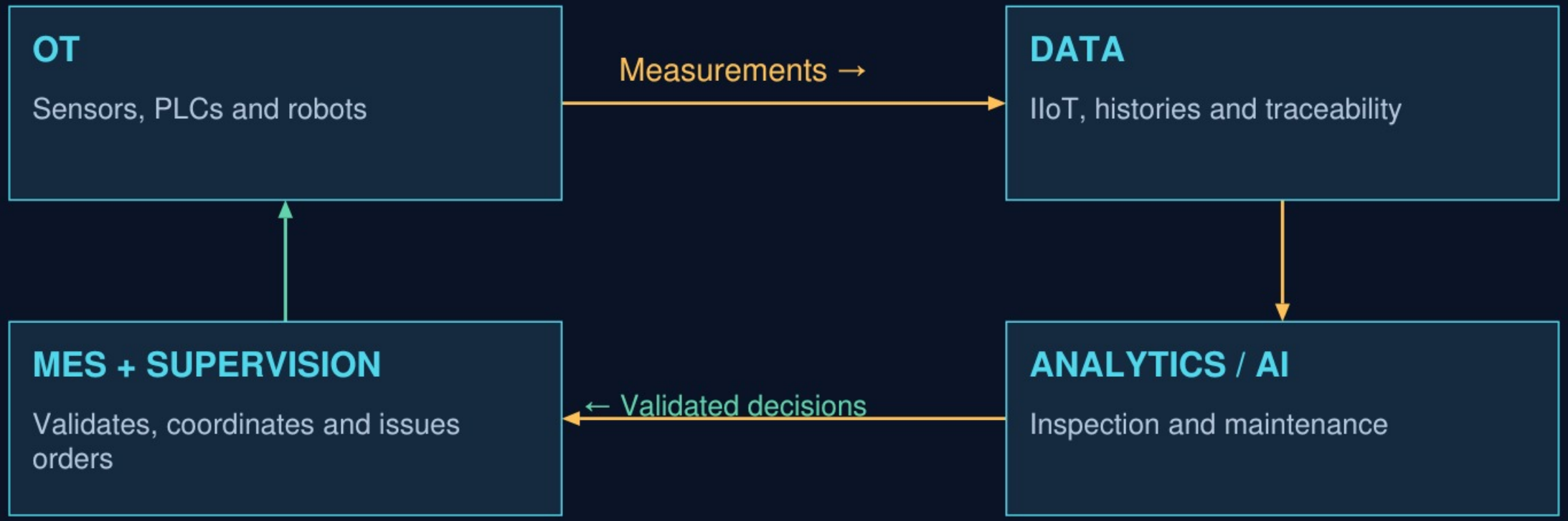
03 · SMART FACTORIES

Automation connected to data and artificial intelligence



Architecture of a smart factory

From sensors to operations management.



Data and orders close the loop; AI requires safe, validated control.

Four technologies transforming the plant

An architecture of complementary capabilities.

01

Robotics

Welding, transport and assembly.

02

AI vision

Surface inspection and defects.

03

Digital twin

Virtual model and scenario tests.

04

Analytics

Failure and bottleneck prediction.

Improvement comes from reducing variability, waits and rework.

Industrial automation: three reference points

Be precise about units and provenance.

354,000

Industrial robots installed in China

2025 · IFR

59%

China’s share of global installations

2025 · IFR

>700

Robots at Xiaomi Auto Factory

Manufacturer description · Xiaomi

Industrial robots ≠ humanoids; annual installations ≠ operating stock.

SOURCE IFR

Takt time, cycle time and throughput

Three frequently confused concepts.

Demand rhythm

- Takt = available time / required units.
- A market and planning reference.
- Defines a synchronisation target.
- May vary with the model mix.

Actual capacity

- Cycle time = time per station operation.
- The bottleneck constrains output.
- Downtime, scrap and buffers affect flow.
- Performance depends on all stages.

The HTML laboratory explores this bottleneck logic.

OEE: availability × performance × quality

Read operating losses without counting them twice.

AVAILABILITY

Operating time / planned production time

×

PERFORMANCE

Ideal cycle × total count / operating time

×

QUALITY

Good count / total count

OEE = A × P × Q

Example: 0.88 × 0.91 × 0.96 = 0.768768 ≈ 76.9%

The three factors multiply; laboratory values are assumptions.

Digital twins: from representation to decision

A simulator needs plant data and validation.

Physical model / data

- Line sensors and material balances.
- Discrete events and operating rules.
- Calibration of queues and durations.
- Validation against historical data.

Experimentation

- Simulate investments and shifts.
- Test supply restrictions.
- Evaluate quality and maintenance.
- Explore risks before changing a plant.

Without factory data, this is a teaching model rather than a predictive digital twin.

Xiaomi Auto: highly integrated manufacturing

Data and capabilities presented by the manufacturer.

01

Campus

718,000 m² and six NEV workshops.

02

Robots

Over 700 robots in automated operations.

03

Die-casting

Large-part cells and process control.

04

Validation

Laboratories, testing and an internal track.

Automation centralises data; overall efficiency still requires measurement.

SOURCE XIAOMI

ECOSYSTEMS

04 · COMPANIES AND VALUE CHAINS

From component control to global competition



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Four complementary corporate strategies

Business models differ.

01

BYD

Battery–vehicle integration and NEV platforms.

02

CATL

Deep specialisation in batteries and energy.

03

Xiaomi

Software, product ecosystem and its own factory.

04

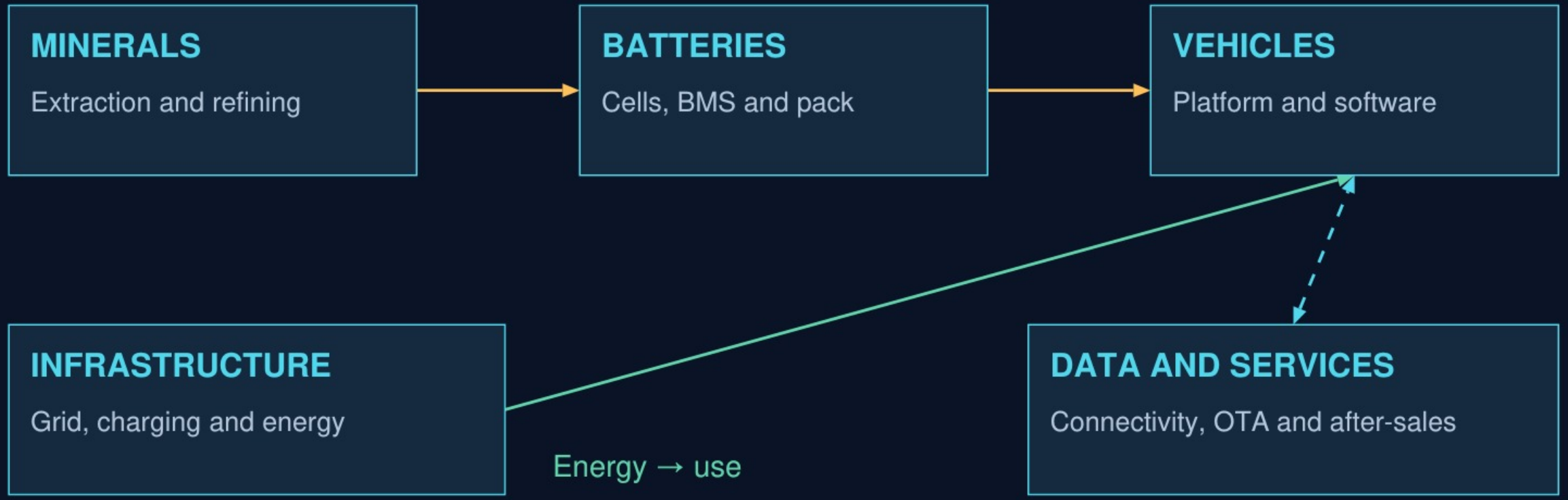
XPeng / Geely

Software, platforms and diverse segments.

Examine supplier networks rather than attributing all innovation to one company.

The value chain becomes a network

Chemistry, electronics, energy and software converge.



Arrows: physical supply (gold), energy (green), information (cyan).

Vertical integration: advantages and risks

An industrial organisation dilemma.

Potential advantages

- Lower coordination costs.
- Design–factory feedback.
- Control of critical supplies.
- Rapid product iteration.

Structural risks

- High capital intensity.
- Managerial and technical complexity.
- Exposure to aggregate demand.
- Less flexibility when standards change.

Optimal integration depends on transaction costs and economies of scope.

Where does AI appear inside the vehicle?

Distinguish driver assistance from automated driving.

01

Perception

Cameras, radar and sensor fusion.

02

Planning

Trajectories, rules and risk assessment.

03

Control

Braking, steering and actuators.

04

Validation

Simulation, safety and field data.

Advanced driver assistance does not imply unsupervised autonomy.

Charging: one infrastructure, several markets

The charger is part of a wider system.



Operations and services support charging infrastructure.

GEOECONOMICS

05 · ECONOMICS AND TRADE

Industrial specialisation, productivity and competitive tensions



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Industrial concentration and resilience

Approximate shares at different stages.

70%

Global electric car production

China · 2025 · IEA

>80%

Battery cell production

China · 2025 · IEA

≈85%

Cathode active material

China · 2025 · IEA

Concentration can lower costs while increasing exposure to common shocks.

SOURCE TRADE

Exporting vehicles is one step in internationalisation

Evaluate localisation, service networks and standards.

Direct exports

- Rapid market entry.
- Lower initial investment.
- Exposure to tariffs and freight costs.
- Cross-border service and parts.

Local production

- Proximity to customers.
- Adjustment to local content requirements.
- Higher CAPEX and installation time.
- Learning with new suppliers.

The choice depends on local scale, logistics, regulation and political risk.

Productivity and value added: a measurement framework

Robot adoption alone does not measure economic performance.

01

TFP

Output growth beyond measured factor contributions.

02

Capacity

Effective plant use and flexibility.

03

Profitability

Margins and returns on capital.

04

Learning

Accumulated improvements in defects, time and quality.

Technical efficiency, economic efficiency and welfare are distinct concepts.

The externalities of electrification

Requires life-cycle assessment and energy context.

Potential benefits

- Lower local urban pollution.
- Less oil used for transport.
- Flexibility to integrate renewables.
- New technological employment.

Costs and trade-offs

- Resource-intensive mining and processing.
- Manufacturing energy use.
- Reuse and end-of-life management.
- Peak electricity demand risks.

The balance depends on electricity mix, mileage, chemistry and recycling.

SIMULATION

06 · DYNAMIC LABORATORY

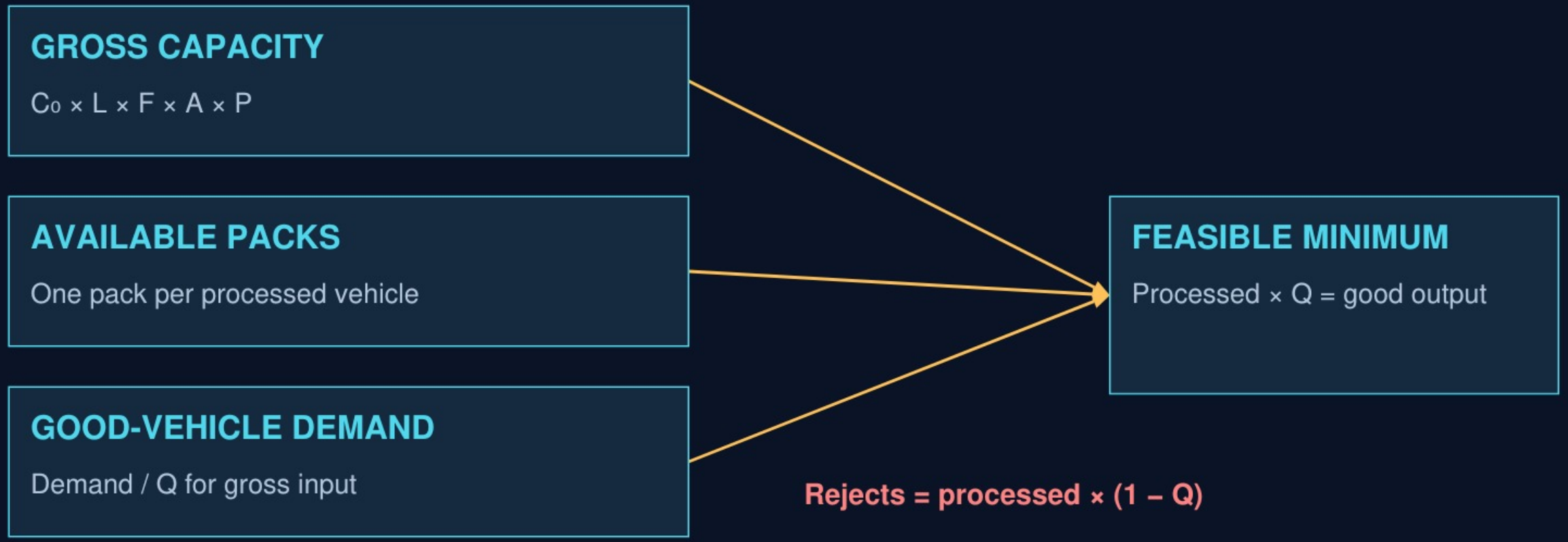
Model productive capacity as a system of constraints



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Conceptual model: output constrained by bottlenecks

Good output under demand and physical capacity limits.



Rejected vehicles also consume packs and operational energy.

Five scenarios in the HTML laboratory

Eight parameters can also be adjusted manually.

REFERENCE

Cap. 18,000; packs 17,000
Demand 19,000

AI + ROBOTS

A 94%; P 96%; Q 98%
Automation 85/100

PACK CONSTRAINT

Packs 9,000; cap. 22,000
Demand 22,000

EV ACCELERATION

Demand 34,000
Cap. 20,000; packs 26,500

INTEGRATED

Cap. and packs 26,500
Q 99%; energy 440

Five presets, eight controls and common equations over 20 quarters.

Three key outputs for discussion

Simulated indices and quantities, rather than empirical data.

units/month

Good plant output

Depends on the bottleneck.

OEE

Availability × performance × quality

Assumed synthetic indicator.

kWh/good

Operational electricity consumption

Including energy used for rejects.

The laboratory generates teaching trajectories, not company forecasts.

SOURCE MODEL

The challenge of calibration and causality

Necessary for serious applied analysis.

For a teaching model

- Explicit assumptions.
- Visible variables and equations.
- Reproducible scenarios.
- Monotonicity tests.

For an operational twin

- Time series of durations and failures.
- Distributions of scrap and cycle times.
- Out-of-sample validation.
- Maintenance, costs and safety.

Conceptual models and validated digital twins serve different objectives.

CONCLUSION

07 · STRATEGIC SYNTHESIS

What China’s experience teaches about future manufacturing



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Five conclusions about industrial transformation

Connecting technology to economic analysis.

SCALE

Market and learning

SYSTEM

Chemistry, hardware and data

COMPETITION

Innovation and margins

RESILIENCE

Concentration and exposure

PRODUCTIVITY

Quality, investment and returns

Evaluation requires technical and economic evidence.

Open questions towards 2030

Conditional scenarios rather than point forecasts.

Technological change

- Will chemistries and platforms converge?
- What happens with sodium-ion batteries?
- When will automated driving scale?
- How does physical AI change the factory?

Political economy

- Where will new production be located?
- Can excess capacity be absorbed?
- How will standards and tariffs evolve?
- What will recycling cost?

Outcomes depend on technology, demand and institutions together.

A route towards a national technology strategy

Useful lessons for Latin America.

01

Diagnosis

Map capabilities and dependencies.

→

02

Specialisation

Choose stages with real advantages.

→

03

Institutions

Training, credit and infrastructure.

→

04

Evaluation

Sustainable productivity and exports.

Industrial policy needs implementation capacity and measurement.

Primary sources and audiovisual guide

Full links are available in speaker notes and the web guide.

IEA IEA, Global EV Outlook 2026

BAT IEA, Electric vehicle batteries, 2026

TRADE IEA, Manufacturing and trade, 2026

CATL CATL, 2025 Annual Report highlights

IFR IFR, World Robotics 2026

BYD BYD, 2025 results / Fortune 2026

XIAOMI Xiaomi Auto Factory

WEF CATL, Smart Manufacturing

SOURCE IEA

SYNTHESIS

From battery to industrial system

Technological advantage combines knowledge, scale, data and organisation.

Sergio Gevatschnaider

